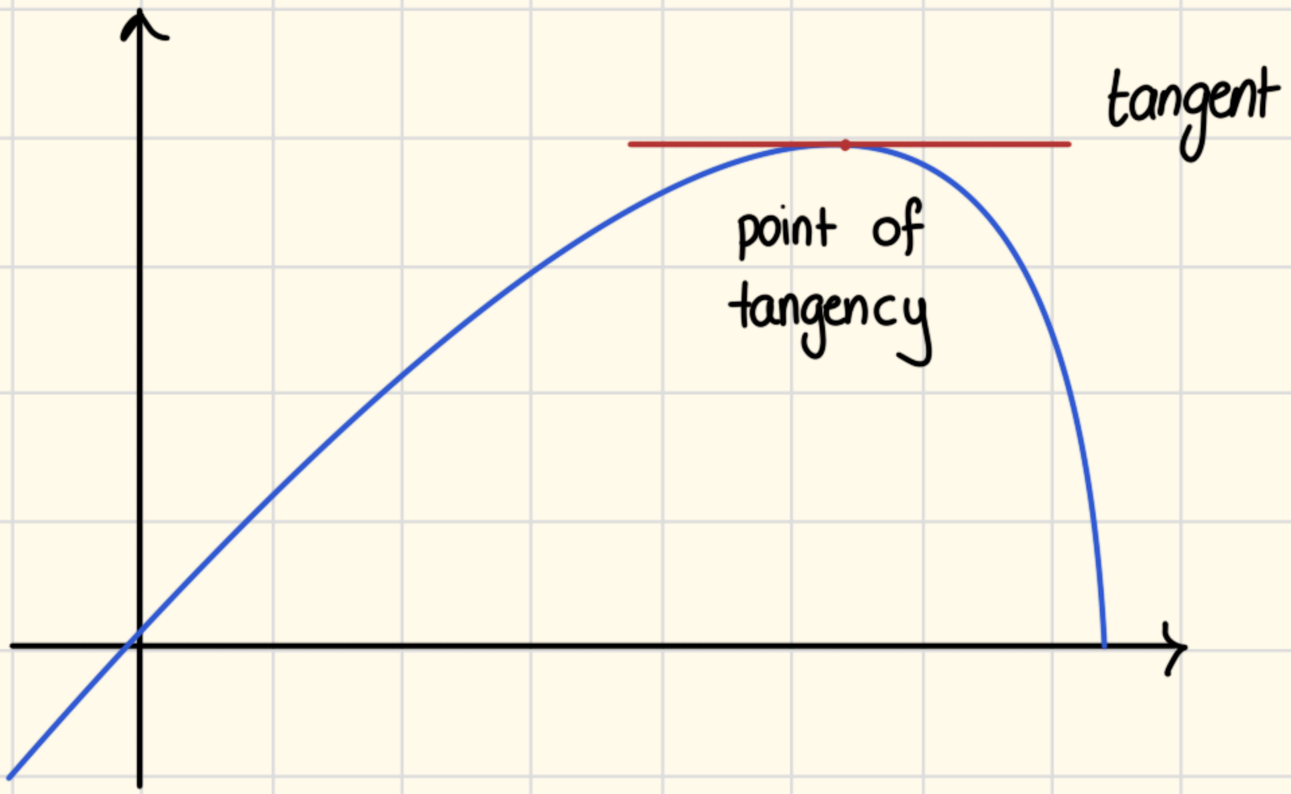
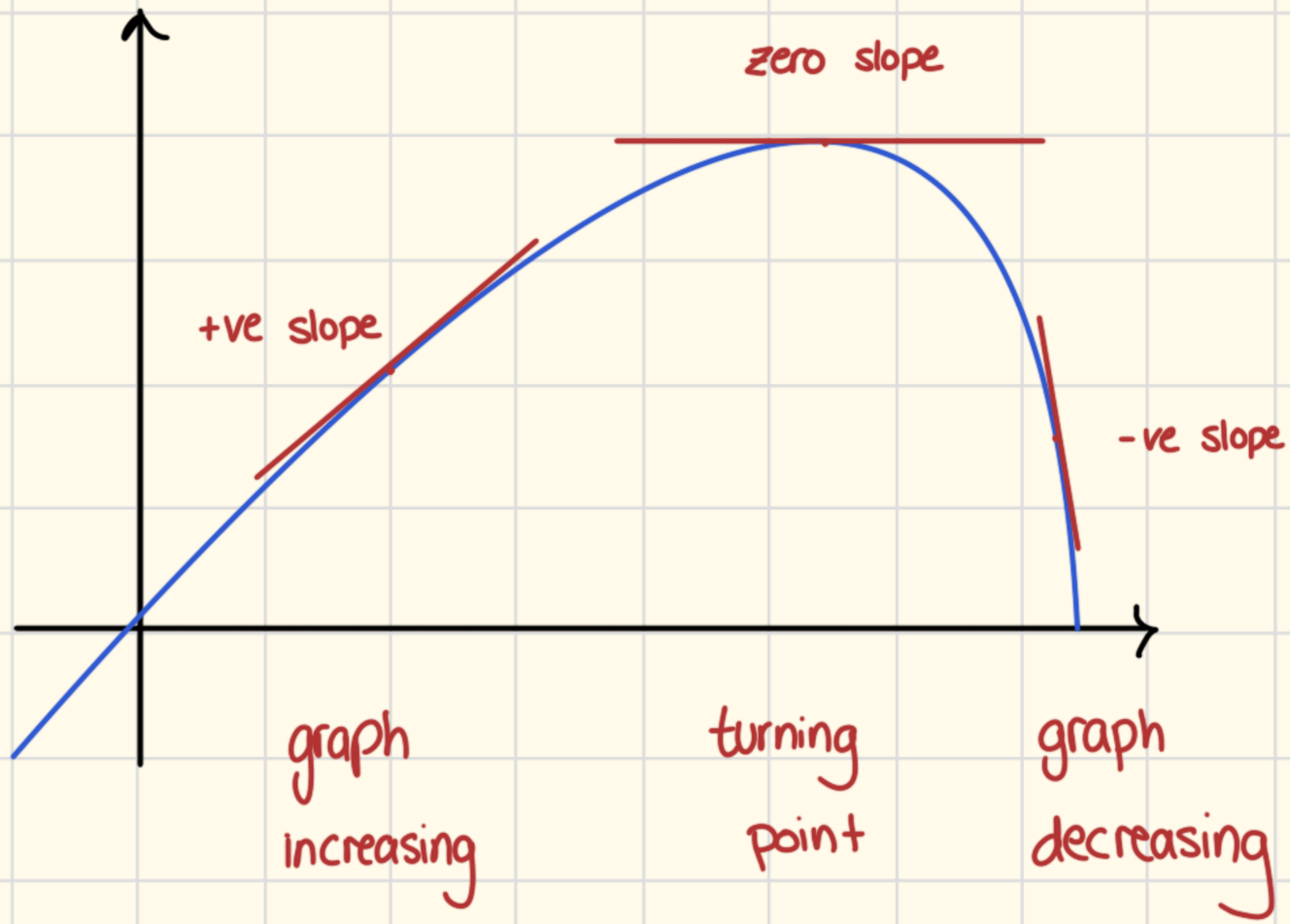
DIFFERENTIATION

A tangent is a line that just touches a curve at a point.



DIFFERENTIATION

Gradients of tangents tell you a lot about a curve.



How can you find the slope of a curve?

DIFFERENTIATION

We can find the gradient of a function $f(x)$

using the gradient function $f'(x)$.

$\frac{dy}{dx}$ OR $f'(x)$

is called the derivative

the operation is called differentiation.

Power Rule

For $y = x^n$, the derivative

$$\frac{dy}{dx} = n x^{n-1}$$

Power Rule

Find $\frac{dy}{dx}$ of

a) $x^5 \rightarrow 5x^4$

b) $3x^5 \rightarrow 15x^2$

c) $3x^5 + \frac{1}{x^2} \rightarrow 15x^2 - \frac{2}{x^3}$

d) $\frac{5}{x^3} \rightarrow -\frac{15}{x^4}$

e) $x^{1/2} \rightarrow \frac{1}{2}x^{-1/2}$

f) $x \rightarrow 1$

g) $6x \rightarrow 6$

h) $5 \rightarrow 0$

i) $5x(x+2) \rightarrow 10x + 10$

j) $\frac{1}{x}(2x^3 + x) \rightarrow 4x$

Equation Of A Line

Find the equation of a tangent line of :

$y = 4x - \frac{1}{x^2}$ at i) $x = 1$ ii) $x = -2$

Solution

We need both the gradient and a point of the tangent line.

Gradient

$$\frac{dy}{dx} = 4 + \frac{2}{x^3}$$

For $x = 1$, $\frac{dy}{dx} = 6$

For $x = -2$, $\frac{dy}{dx} = \frac{15}{4}$

Point

$$y = 4x - \frac{1}{x^2}$$

For $x = 1$, $y = 3$

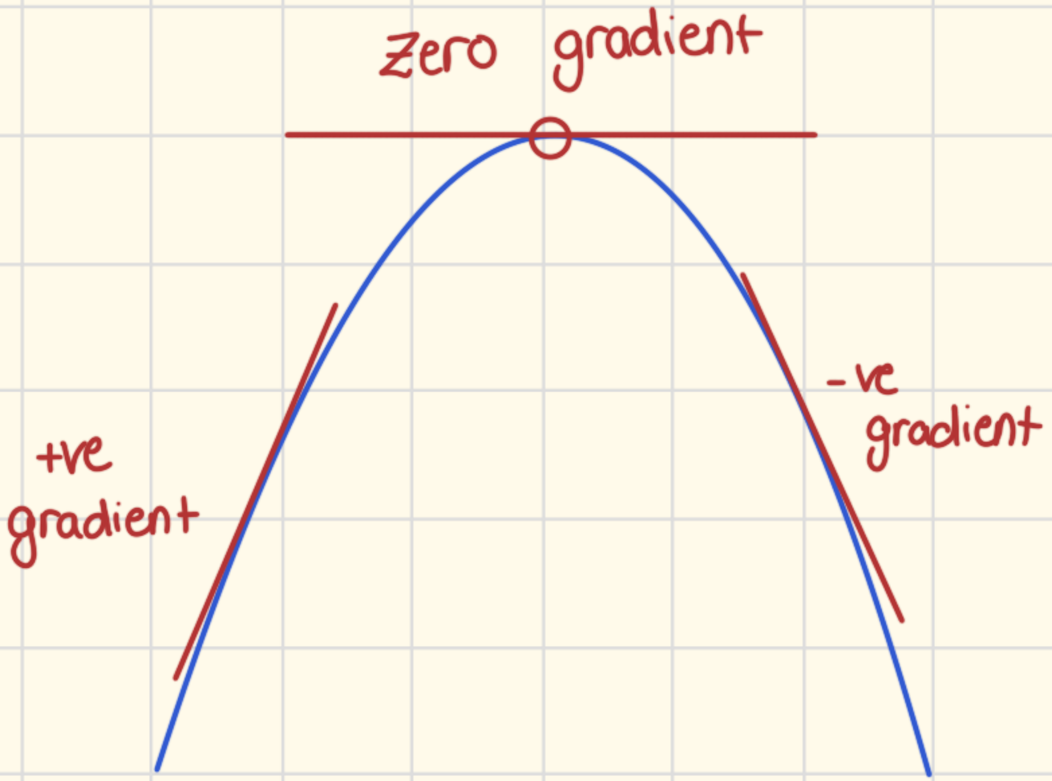
For $x = -2$, $y = -\frac{33}{4}$

Equation

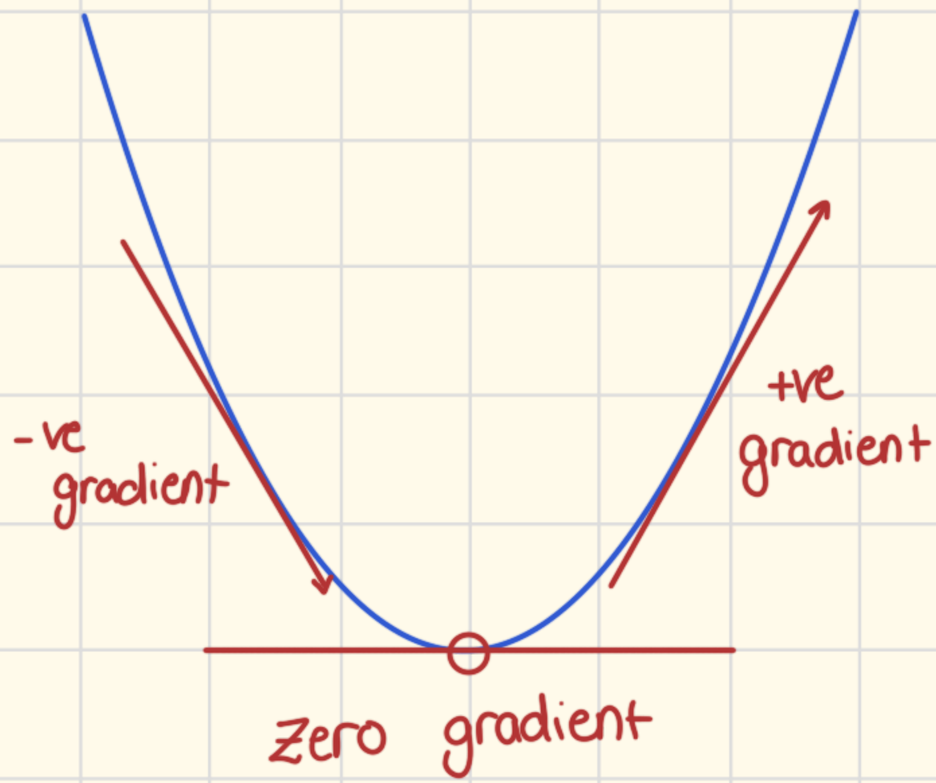
$$y - 3 = 6(x - 1)$$

$$y + \frac{33}{4} = \frac{15}{4}(x + 2)$$

Maximum And Minimum Points



max



min

Turning points are points with zero gradient

OR $\frac{dy}{dx} = 0$.